INFO8006 Introduction to Artificial Intelligence

Exercise session 5: Reasoning over time

Recap

Markov models

The world is a discrete series of time slices, where $\mathbf{X}_t$ denotes the set of *unobservable* state variables and $\mathbf{E}_t$ the set of *observable* evidence variables at time t. We specify a prior $\mathbf{P}(\mathbf{X}_0)$, a *transition model* $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1})$ and a *sensor model* $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{0:t-1})$, under

- the first-order *Markov assumption* $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1}) = \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{t-1})$, which defines a first-order *Markov process*;
- the *sensor Markov assumption* $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{0:t-1}) = \mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$;
- the *stationarity assumption*: the transition and sensor models are the same for all t.

The joint distribution of all variables up to t is then

$$\mathbf{P}(\mathbf{X}_{0:t}, \mathbf{E}_{1:t}) = \mathbf{P}(\mathbf{X}_0) \prod_{i=1}^{t} \mathbf{P}(\mathbf{X}_i \mid \mathbf{X}_{i-1}) \mathbf{P}(\mathbf{E}_i \mid \mathbf{X}_i).$$

Inference tasks

Given the evidence $\mathbf{e}_{1:t}$ collected up to time t, we define

- *prediction*: $\mathbf{P}(\mathbf{X}_{t+k} \mid \mathbf{e}_{1:t})$ for $k > 0$;
- *filtering*: $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$, the *belief state* about the current hidden state $\mathbf{X}_t$;
- *smoothing*: $\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t})$ for $0 \leq k < t$;
- *most likely explanation*: $\arg\max_{\mathbf{x}_{1:t}} P(\mathbf{x}_{1:t} \mid \mathbf{e}_{1:t})$.

Prediction

To predict the future, the belief state is pushed through the transition model,

$$\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}) = \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t \mid \mathbf{e}_{1:t}),$$

and this is repeated up to $t + k$.

Filtering

The *Bayes filter* maintains the belief state $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ as time passes and new evidence $\mathbf{e}_{t+1}$ is collected. It alternates between a predict step, which projects the current belief state forward from t to $t + 1$ through the transition model, and an update step, which updates this new state using the evidence $\mathbf{e}_{t+1}$:

$$\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1}) \propto \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t \mid \mathbf{e}_{1:t}).$$

The belief state is a *forward message* $\mathbf{f}_{1:t}$, with $\mathbf{f}_{1:t+1} \propto \text{forward}(\mathbf{f}_{1:t}, \mathbf{e}_{t+1})$. For continuous variables, sums are replaced with integrals and probability mass functions with densities:

$$p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t+1}) \propto p(\mathbf{e}_{t+1} \mid \mathbf{x}_{t+1}) \int p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) p(\mathbf{x}_t \mid \mathbf{e}_{1:t}) \, \mathrm{d}\mathbf{x}_t.$$

Smoothing

Dividing the evidence $\mathbf{e}_{1:t}$ into $\mathbf{e}_{1:k}$ and $\mathbf{e}_{k+1:t}$,

$$\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t}) \propto \mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k) = \alpha \, \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:t},$$

where $\times$ is a pointwise multiplication of vectors and the *backward message* $\mathbf{b}_{k+1:t} = \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k)$ is computed by backwards recursion, $\mathbf{b}_{k+1:t} = \text{backward}(\mathbf{b}_{k+2:t}, \mathbf{e}_{k+1})$.

Most likely explanation

The *Viterbi algorithm* replaces the forward message with $\mathbf{m}_{1:t} = \max_{\mathbf{x}_{1:t-1}} \mathbf{P}(\mathbf{x}_{1:t-1}, \mathbf{X}_t \mid \mathbf{e}_{1:t})$, the probability of the most likely path to each state, computed as

$$\mathbf{m}_{1:t+1} \propto \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \max_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) \mathbf{m}_{1:t}.$$

Hidden Markov models

A *hidden Markov model* (HMM) is a Markov process in which the state $\mathbf{X}_t = X_t$ and the evidence $\mathbf{E}_t = E_t$ are both single discrete random variables, with domains $D_{X_t} = \{1, \ldots, S\}$ and $D_{E_t} = \{1, \ldots, R\}$. The forward-backward algorithm then reduces to matrix-vector operations:

- the prior $\mathbf{P}(X_0)$ becomes a column vector $\mathbf{f}_0 \in \mathbb{R}_+^S$;
- the transition model $\mathbf{P}(X_t \mid X_{t-1})$ becomes an $S \times S$ *transition matrix* $\mathbf{T}$ such that $\mathbf{T}_{ij} = P(X_t = j \mid X_{t-1} = i)$;
- the sensor model $\mathbf{P}(E_t \mid X_t)$ is an $S \times R$ *sensor matrix* $\mathbf{B}$ such that $\mathbf{B}_{ij} = P(E_t = j \mid X_t = i)$;
- the *observation matrix* $\mathbf{O}_t$ is a diagonal matrix whose elements correspond to the column e_t of $\mathbf{B}$.

With column vectors for the forward and backward messages,

$$\mathbf{f}_{1:t+1} = \alpha \mathbf{O}_{t+1} \mathbf{T}^T \mathbf{f}_{1:t},$$
$$\mathbf{b}_{k+1:t} = \mathbf{T} \mathbf{O}_{k+1} \mathbf{b}_{k+2:t},$$

where $\mathbf{b}_{t+1:t}$ is an all-one vector of size S.

Stationary distribution (optional)

Without new evidence, the predicted distribution may converge, as k grows, to a *stationary distribution*: a fixed point $\mathbf{P}(\mathbf{X}_{t+1}) = \mathbf{P}(\mathbf{X}_t)$ of the predict step $\mathbf{P}(\mathbf{X}_{t+1}) = \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t)$. For an HMM, it is a vector $\mathbf{f}$ such that $\mathbf{f} = \mathbf{T}^T \mathbf{f}$.

Matrix multiplication

In index notation,

$$\mathbf{C} = \mathbf{A}\mathbf{B} \quad \Leftrightarrow \quad \mathbf{C}_{ij} = \sum_u \mathbf{A}_{iu} \mathbf{B}_{uj} \quad \forall \, i, j,$$
$$\mathbf{d} = \mathbf{A}\mathbf{f} \quad \Leftrightarrow \quad \mathbf{d}_i = \sum_u \mathbf{A}_{iu} \mathbf{f}_u \quad \forall \, i.$$

Cheat sheet for Gaussian distributions

If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\mathbf{a}\\ \mathbf{b}\end{pmatrix}, \begin{pmatrix}\mathbf{A} & \mathbf{C}\\ \mathbf{C}^T & \mathbf{B}\end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a}, \mathbf{A}),\\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b}, \mathbf{B}),\\ p(\mathbf{x} \mid \mathbf{y}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a} + \mathbf{C}\mathbf{B}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{C}\mathbf{B}^{-1}\mathbf{C}^T),\\ p(\mathbf{y} \mid \mathbf{x}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}). \end{aligned}$$

If $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \mathbf{m}, \mathbf{P})$ and $p(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(\mathbf{y} \mid \mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R})$, then the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is Gaussian with

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\mathbf{m}\\ \mathbf{H}\mathbf{m} + \mathbf{u}\end{pmatrix}, \begin{pmatrix}\mathbf{P} & \mathbf{P}\mathbf{H}^T\\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R}\end{pmatrix}\right).$$

Kalman filter

The *Kalman filter* is a special case of the Bayes filter for continuous state and evidence variables, which assumes a Gaussian prior $p(\mathbf{x}_0)$, a linear Gaussian transition model and a linear Gaussian sensor model,

$$\begin{aligned} p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t+1} \mid \mathbf{A}\mathbf{x}_t + \mathbf{b}, \mathbf{Q}),\\ p(\mathbf{e}_t \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{e}_t \mid \mathbf{H}\mathbf{x}_t + \mathbf{d}, \mathbf{R}). \end{aligned}$$

Under these assumptions, the predict and update steps both yield Gaussian distributions, $p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t}) = \mathcal{N}(\mathbf{x}_{t+1} \mid \boldsymbol{\mu}^-_{t+1}, \boldsymbol{\Sigma}^-_{t+1})$ and $p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t+1}) = \mathcal{N}(\mathbf{x}_{t+1} \mid \boldsymbol{\mu}_{t+1}, \boldsymbol{\Sigma}_{t+1})$, whose parameters are derived in closed form with the Gaussian identities above.

Exercises

In session: exercises 1, 2.

Exercise 1 Load of a shared server (Exam, January 2023)

A scheduling program submits jobs to a shared compute server, whose load varies over time. Each job runs under either a heavy or a light load. From past logs, the program knows that if a job runs under a heavy load, there is a 40% chance that the next job also runs under a heavy load. If a job runs under a light load, there is a 70% chance that the next job also runs under a light load. The program cannot see the load, but it sees the response time of each job, which is either fast, normal or slow. If the load is heavy, the response time is fast with probability 20% and slow with probability 60%. If the load is light, it is fast with probability 50% and slow with probability 30%.

(a) Define the components of a hidden Markov model (HMM), where the state $X_t \in \{\texttt{heavy}, \texttt{light}\}$ is the load under which the t-th job runs and the evidence $E_t \in \{\texttt{fast}, \texttt{normal}, \texttt{slow}\}$ is its response time.

(b) At some point, the server stops reporting the response time of every other job (the 2nd, the 4th, ...), so that the program only observes the 1st, 3rd, 5th, ... jobs. Sketch a Bayesian network describing this new process.

(c) Express, in terms of the HMM components, the distribution $P(X_t \mid e_{1:t:2})$ of the load for the t-th job given the observations $e_{1:t:2}$ of every other job. Note that, $e_{1:6:2} = (e_1, e_3, e_5)$ and $e_{1:7:2} = (e_1, e_3, e_5, e_7)$. Separate the cases where t is even and odd.

(d) Given the observations $e_{1:3:2} = (e_1, e_3) = (\texttt{slow}, \texttt{fast})$, compute the probability that the 3rd job runs under a heavy load.

Exercise 2 Umbrella world (AIMA ch. 14)

A security robot guards a secret underground installation. It would like to know whether it is raining today, but its only access to the outside world occurs each morning, when it sees the director coming in with, or without, an umbrella. For each day t, the evidence is a single variable $\mathit{Umbrella}_t \in \{1, 0\}$, *i.e.* whether the umbrella appears or not, and the (hidden) state is a single variable $\mathit{Rain}_t \in \{1, 0\}$, *i.e.* whether it is raining or not.

The robot's model assumes that from one day $t-1$ to the next t, the chances that the weather stays the same are 70 %. It also assumes that the director brings an umbrella 90 % of the time when it is raining, and 20 % of the time otherwise.

(a) The robot represents the umbrella world as a Markov model. What formal assumptions correspond to these beliefs?

(b) Sketch a Bayesian network structure describing the umbrella world and provide the transition and sensor models.

(c) Express the distributions $P(R_{t+1} \mid R_{t-1})$, $P(U_t \mid R_{t-1})$ and $P(R_t \mid R_{t-1}, U_t)$ in terms of the transition and sensor models.

(d) Suppose the robot observes an unending sequence of days on which the umbrella appears. Show that, as the days go by, the probability of rain on the current day tends monotonically towards a fixed point. Calculate this fixed point.

(e) Now consider forecasting further and further into the future, given t umbrella observations. Is there a fixed point? If yes, compute its exact value.

Exercise 3 Coin-tossing machine

The surveillance program of a casino watches a coin-tossing machine. The machine holds 3 biased coins a, b and c, whose probabilities of heads are 80 %, 50 % and 20 %, respectively. It first picks one of the coins at random. Then, 4 times, it either keeps the current coin with probability $\frac{2}{3}$ or replaces it by one of the two other coins, chosen at random, then tosses the coin and shows the result. The program only sees the results: heads, heads, tails, heads.

(a) Provide a hidden Markov model (HMM) that describes the process.

(b) What are the probabilities of the last coin given the sequence of results?

(c) What are the probabilities of the coin picked first given the sequence of results? And of the first coin tossed?

(d) What is the most likely sequence of tossed coins?

Exercise 4 Ball on a spring

The "Foire de Liège" has a new attraction called "Super Spring Ultra Pro Max XXL" which consists in a ball attached to a spring on a platform. The participants take seat in the ball locked at some position. The ball is then released and pulled by the spring which makes it oscillate back and forth. After a few seconds, the ball is stopped by magnetic brakes. You notice that the final position of the ball is different each time. As the ball is opaque, you wonder if it is possible for the participants to guess where the ball stopped, given their perception of acceleration.

You more or less remember your Newtonian mechanics class and model the movement of the ball as a series of transitions

$$\begin{aligned} p_t &= p_{t-1} + \Delta t\, \dot{p}_{t-1} + \frac{1}{2}\Delta t^2\, \ddot{p}_{t-1} \\ \dot{p}_t &= \dot{p}_{t-1} + \Delta t\, \ddot{p}_{t-1} \\ \ddot{p}_t &= \rho w_{t-1} - \kappa\, p_{t-1} - \eta\, \dot{p}_{t-1} \\ w_t \mid w_{t-1} &\sim \mathcal{N}(\alpha w_{t-1}, \sigma_w^2) \end{aligned}$$

where p_t, $\dot{p}_t$ and $\ddot{p}_t$ are respectively the position, velocity and acceleration of the ball at timestep t, w_t is the wind at timestep t, Δt is the time elapsed between $t-1$ and t, ρ is the thrust coefficient of the wind, κ is the stiffness of the spring, η is the friction coefficient of the platform and α is the persistence of the wind. You estimate that the ball starts $10.0 \pm 0.5\,\mathrm{m}$ at the left of the spring with a negligible speed $0.0 \pm 0.1\,\mathrm{m\,s^{-1}}$ and acceleration $0.0 \pm 0.1\,\mathrm{m\,s^{-2}}$. You assume that the wind has a stationary distribution. You assume that the human perception of acceleration follows an unbiased Gaussian distribution. You also assume that a good model to describe the uncertainty in your initial guess of the ball's state is a Gaussian model, with the standard deviation given by your estimated bounds.

(a) You wish to predict the state of the ball given the perceptions of a participant. Define the components of a Kalman filter in this context.

(b) Express the distribution $p(x_t \mid e_{1:t})$ with respect to the components defined previously. Use the cheat sheet for Gaussian models.

(c) Represent the transition and sensor models as a dynamic Bayesian network.

Exercise 5 Robot state estimation (Exam, January 2024)

Let us consider a continuous first-order Markov process describing the motion of a robot in a one-dimensional world. On average, the robot moves forward by a distance b at each timestep. The robot's position x_t at time t is not directly observable, but can be estimated using a sensor that measures the distance between the robot and the sensor. Assuming,

- the sensor is fixed at position 0,

- a Gaussian prior $\mathcal{N}(x_0 \mid \mu_0, \sigma_0^2)$ on the initial position of the robot,
- a Gaussian transition model $\mathcal{N}(x_{t+1} \mid x_t + b, \sigma_x^2)$,
- a Gaussian observation model $\mathcal{N}(e_t \mid x_t, \sigma_e^2)$,

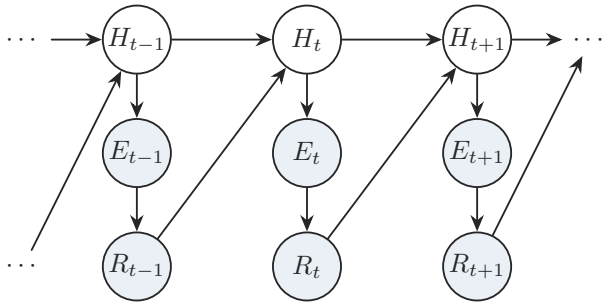
derive the update equations of the Bayes filter. More specifically, since the belief distribution is Gaussian, derive the update equations of its mean μ_{t+1} and variance σ_{t+1}^2.

Do the update equations make sense in the following limiting cases? Use your physical intuition.

- The sensor is extremely noisy.
- The sensor is perfect.
- The transition model is extremely noisy.
- The transition model is perfect.

Exercise 6 Human-robot interaction (CS188, Fall 2017)

In the near future, autonomous robots will live among us. Therefore, the robots need to know how to act appropriately in the presence of humans. In this question, we explore a simplified model of this interaction. We assume that we can observe the robot's actions at time t, R_t, and an evidence observation, E_t, directly caused by the (hidden) human state, H_t. Robot actions from the current time-step affect the human state in the next-time step, as illustrated in the Bayesian network given below.



Assuming discrete random variables and given the probability tables $P(H_0)$, $P(E_t \mid H_t)$, $P(R_t \mid E_t)$ and $P(H_{t+1} \mid H_t, R_t)$, your goal is to derive a procedure to maintain a belief $P(H_t \mid e_{1:t}, r_{1:t})$ about the state of the human at time t.

(a) Derive an update equation for incorporating a pair of observations (e_t, r_t) to a given belief state $P(H_t)$.

(b) Derive an update equation for predicting the future state H_{t+1} of the human at time $t+1$ given a belief state $P(H_t \mid e_{1:t}, r_{1:t})$.

(c) Combine both equations to derive a recursive update equation of the belief state $P(H_t \mid e_{1:t}, r_{1:t})$, as observations are collected and time passes.

(d) Let us now assume that all variables are continuous. Discuss how you would compute or approximate the belief state on a computer.

Exercise 7 Server cooling (Exam, August 2019) (AIMA ch. 14)
A monitoring program wants to know whether a server is cooled enough. Each day, the program checks whether the server raised a temperature alert, and whether it throttled its processors. The program makes the following hypotheses:

- The prior probability that the cooling is enough, with no observations, is 0.7.
- The probability that the cooling is enough on day t is 0.8 given that it was enough on the previous day, and 0.3 otherwise.
- The probability of a temperature alert is 0.2 if the cooling is enough, and 0.7 otherwise.
- The probability of throttling is 0.1 if the cooling is enough, and 0.3 otherwise.

(a) Formulate the environment and hypotheses as a dynamic Bayesian network that the program could use to detect insufficient cooling from a sequence of observations. Provide the associated probability tables.

(b) Reformulate the dynamic Bayesian network as a hidden Markov model that has only a single observation variable. Give the complete probability tables for the model.

(c) For the sequence $e_{1:3}$ of observations "no alert, no throttling", "alert, no throttling" and "alert, throttling", calculate the distributions $P(\mathit{Cooled}_t \mid e_{1:t})$ and $P(\mathit{Cooled}_t \mid e_{1:3})$ for $t \in \{1, 2, 3\}$.

Quiz

1. The normal distribution $\mathcal{N}(\mu, \sigma^2)$ is described by the density function
   - ◯ $p(x) = \frac{1}{\sigma} \exp\left(-(z + \exp(-z))\right)$, with $z = \frac{x-\mu}{\sigma}$.
   - ◯ $p(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x-\mu|}{\sigma}\right)$.
   - ◯ $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
   - ◯ $p(x) = \dfrac{1}{\pi\sigma\left[1 + \left(\frac{x-\mu}{\sigma}\right)^2\right]}$.
2. The Kalman filter requires the specification of ...
   - ◯ a prior $\mathbf{P}(\mathbf{X}_0)$ and a transition model $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t)$.
   - ◯ a transition model $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t)$ and an observation model $\mathbf{P}(\mathbf{E}_t \mid \mathbf{x}_t)$.
   - ◯ a prior $\mathbf{P}(\mathbf{X}_0)$, a transition model $P(\mathbf{X}_{t+1} \mid \mathbf{x}_t)$, and an observation model $P(\mathbf{E}_t \mid \mathbf{x}_t)$.
   - ◯ a prior $\mathbf{P}(\mathbf{X}_0)$, a prior $\mathbf{P}(\mathbf{E}_0)$, a transition model $P(\mathbf{X}_{t+1} \mid \mathbf{x}_t)$, and an observation model $P(\mathbf{E}_t \mid \mathbf{x}_t)$.
3. In filtering, the belief state is updated at each timestep using ...
   - ◯ the current belief and the transition model.
   - ◯ the current belief, the observation model and the new evidence.
   - ◯ the transition model, the observation model and the new evidence.
   - ◯ the current belief, the transition model, the observation model and the new evidence.

4. Let us assume a Markov process with hidden state variables $\mathbf{X}_t$, evidence variables $\mathbf{E}_t$, transition model $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{X}_t)$ and sensor model $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$. Which of the following is true?

   ◯ The Bayes filter estimates $\mathbf{P}(\mathbf{X}_{1:t} \mid \mathbf{e}_{1:t})$.

   ◯ Smoothing consists in computing $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1})$ by pushing the belief state $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ forward through the transition model and then updating this new belief state with the evidence $\mathbf{e}_{t+1}$.

   ◯ The Kalman filter computes $\mathbf{P}(\mathbf{X}_{t+k} \mid \mathbf{e}_{1:t})$ by pushing the belief state $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ forward k times through the transition model.

   ◯ The Viterbi algorithm computes the most likely sequence of state values $\mathbf{x}_{1:t}$ given the evidence $\mathbf{e}_{1:t}$.